Which Residual-Stream Directions Are Compositional? A Direction-Level Map of Linear Additivity in Instruction Steering

Ari Holtzman and NeuriCo

Abstract

Activation steering methods often assume that residual-stream directions compose by vector addition, but this assumption has not been mapped systematically at the direction-pair level. We examine which instruction-derived directions are compositional and where composition breaks down. Using Qwen2.5-0.5B-Instruct, we extract 20 instruction directions from IFEval-derived data and evaluate composition with three tests: representation-space additivity on multi-instruction prompts, behavior-space additivity under activation steering, and transfer locality on TruthfulQA and HellaSwag contexts.

We find partial compositionality rather than a global linear subspace. In representation space, additive predictions are moderately aligned with observed displacements (mean cosine 0.518, 95% bootstrap CI [0.478, 0.543]) but poorly calibrated in magnitude (mean R2-like score -0.495; mean relative norm error 0.256). Pair quality varies widely, including negative examples. Related instruction families show slightly higher mean cosine than unrelated families (0.571 vs. 0.509), but this gap is not significant (Mann-Whitney $p = 0.664$). In behavior space, summed steering does not improve joint constraint satisfaction over the best single vector (gain 0.000). In transfer locality tests with synthetic instruction appends, additivity remains high (TruthfulQA 0.868; HellaSwag 0.919).

These results support a mixed regime: local additive structure is real, but global compositionality is uneven and operationally fragile.

1 Introduction

Activation interventions in large language models are increasingly used for controllability, safety, and interpretability. A central assumption behind many methods is simple: if a direction controls attribute A and another controls attribute B, then adding both directions should reliably control A and B together. Our main question is direct: can we treat residual-stream directions as a compositional linear space?

This question matters because steering methods are now used in settings where failures are costly. If direction arithmetic only works in narrow regions, then linear interventions can look reliable on average while failing on specific instruction pairs or contexts. That failure mode is especially concerning for multi-constraint control.

Prior work demonstrates that activation additions can steer behavior and that some vectors compose in practice [Turner et al., 2023, Panickssery et al., 2024, Todd et al., 2024, Stolfo et al., 2025]. Recent work on context-dependent steering suggests that static vectors may capture only local structure [Li et al., 2026]. However, existing evaluations mostly report aggregate gains and do not provide a direction-level map of where composition holds or breaks.

We address this gap with a systematic compositionality study using instruction-derived residual directions. We estimate 20 directions from IFEval-derived prompts and evaluate additivity in three complementary ways: (i) representation-space reconstruction on multi-instruction prompts, (ii) behavior-space composition under causal steering, and (iii) transfer/locality checks on TruthfulQA and HellaSwag.

Our quantitative preview is mixed. Representation additivity is moderate on average (mean cosine 0.518), but reconstruction quality is poor (mean R2-like score -0.495), and some pairs are strongly non-compositional (including negative cosine pairs). Behaviorally, summed steering provides no gain over the best single direction in our checkable subset (gain 0.000). In transfer tests with local synthetic perturbations, additivity is high (0.868–0.919), indicating a locality effect rather than global linearity.

In summary, our main contributions are:

- We construct a direction-level compositionality map for 20 instruction vectors and 85 direction pairs using residual activations from a modern open model.
- We conduct aligned representation-space and behavior-space tests, showing that geometric additivity does not consistently translate into practical compositional steering.
- We quantify statistical uncertainty with bootstrap intervals and nonparametric testing, including a related-vs-unrelated family analysis.
- We provide transfer/locality evidence showing that high local additivity can coexist with weak global composition for full instruction bundles.

Paper organization. Section 2 situates our study in activation steering and compositional representation literature. Section 3 details datasets, direction extraction, and evaluation protocols. Section 4 reports quantitative findings and statistical tests. Section 5 interprets implications and limitations, and section 6 concludes.

2 Related Work

Activation steering with static directions. ActAdd-style methods show that residual-stream additions can causally shift model behavior without retraining [Turner et al., 2023]. CAA extends this idea using contrastive prompt pairs and demonstrates robust control on several tasks [Panickssery et al., 2024]. Instruction-focused steering studies further show gains on constraint-following benchmarks, including multi-instruction settings [Stolfo et al., 2025]. These results establish that linear interventions can work. Unlike these studies, we focus on whether the underlying direction geometry is compositional at the pair level rather than only measuring end-task average improvements.

Function vectors and mechanistic evidence. Function vectors identify causal task signals in model internals and report partial compositional behavior for some task combinations [Todd et al., 2024]. This line of work motivates our representation-level diagnostics. Our analysis complements it by testing instruction families derived from real constrained prompts and by explicitly quantifying reconstruction error, not only alignment.

Beyond static linearity: dynamic and local control. Recent methods propose adaptive weighting or generated vectors to improve multi-attribute control under context shift [Scalena et al., 2024, Sun and collaborators, 2025]. Steering vector-field approaches model context-dependent local directions and report stronger reliability than fixed vectors [Li et al., 2026]. Our findings are consistent with this transition: local additive structure appears strong in transfer perturbation tests, while global composition over full instruction bundles is uneven.

Representation factorization and compositionality. Theory and synthetic studies suggest transformers can learn partially factored internal representations, but factorization quality depends on data and dependencies [Shai and collaborators, 2026]. Mechanistic progress metrics also emphasize that internal structure can be stage-dependent and non-uniform [Nanda et al., 2023]. We contribute empirical evidence in a practical steering setup, showing a mixed regime where some direction pairs are additive and others interfere substantially.

3 Methodology

3.1 Problem Setup

We study whether instruction directions in a transformer residual stream are compositionally additive. For a prompt x, let $h_\ell(x)$ be the last-token residual activation at layer ℓ. For instruction i, we estimate a direction $\boldsymbol{d}_i$ from paired constrained/de-constrained prompts. For a multi-instruction prompt with set S, we compare observed displacement and additive prediction:

$$\Delta_{\text{obs}}(x, S) = h_\ell(x_{\text{constrained}}) - h_\ell(x_{\text{base}}), \tag{1}$$

$$\Delta_{\text{pred}}(S) = \sum_{i \in S} \boldsymbol{d}_i. \tag{2}$$

Our core question is whether Δ_{pred} is a good approximation to Δ_{obs} across pairs and contexts.

3.2 Model, Data, and Direction Extraction

Model and implementation. We use QWEN2.5-0.5B-INSTRUCT with 'torch 2.5.1+cu124', 'transformers 5.5.0', 'datasets 4.8.4', 'scipy 1.17.1', and 'pandas 3.0.2'. Experiments run on two RTX 3090 GPUs (24GB each). We fix seed 42 and intervene at middle layer 12 of 24.

Datasets. Direction discovery and in-domain evaluation use 541 IFEval-derived examples from `ifeval_wo_instructions.jsonl`. Transfer checks use TruthfulQA validation (817 prompts) and HellaSwag validation (10,042 prompts).

Direction estimation. For each instruction ID with at least 10 single-instruction examples, we compute per-example residual differences between constrained and de-constrained forms and take the mean. This yields 20 instruction directions.

3.3 Evaluation Protocols

Experiment 1: representation compositionality. On multi-instruction prompts, we evaluate additivity with:

- cosine alignment: $\cos(\Delta_{\text{obs}}, \Delta_{\text{pred}})$,
- relative norm error: $|\|\Delta_{\text{pred}}\| - \|\Delta_{\text{obs}}\||/\|\Delta_{\text{obs}}\|$,
- R2-like reconstruction score against the observed displacement baseline.

Experiment 2: behavioral compositionality. For exactly two checkable instructions per prompt, we generate under four conditions: none, A, B, and $A{+}B$. Automatic rule-based checkers score joint satisfaction. We report gain over the best single condition.

Experiment 3: transfer/locality. On TruthfulQA and HellaSwag prompts, we append synthetic instruction snippets and test local additivity:

$$\Delta_{\text{obs}} = h(x + i + j) - h(x), \tag{3}$$

$$\Delta_{\text{pred}} = [h(x + i) - h(x)] + [h(x + j) - h(x)]. \tag{4}$$

We measure cosine alignment between these two vectors.

3.4 Baselines and Statistical Analysis

Baselines include no steering, single-direction steering, and best-single oracle comparison for additive steering. We compute 95% bootstrap confidence intervals for key means and use Mann-Whitney U for related-vs-unrelated pair comparison. We also run a full rerun with the same seed and compare summary metrics for determinism.

4 Results

4.1 Main Quantitative Findings

What is the headline result? As summarized in table 1, representation-space additivity is only partially reliable. Mean cosine is 0.518 with a tight 95% bootstrap interval [0.478, 0.543], but

Metric Group	Metric	Value	Interpretation
Setup	Selected directions	20	Instruction IDs with at least 10 single-instruction samples
Setup	Multi-instruction examples	174	Evaluation set for representation compositionality
Representation	Mean cosine alignment	**0.518**	Moderate directional agreement
Representation	Mean R2-like score	-0.495	Poor reconstruction quality
Representation	Mean relative norm error	0.256	Non-trivial magnitude mismatch
Representation	95% bootstrap CI (cosine)	[0.478, 0.543]	Stable but clearly sub-ceiling
Pair families	Related-pair mean cosine	0.571	Slightly higher than unrelated
Pair families	Unrelated-pair mean cosine	0.509	Lower average compositionality
Pair families	Mann-Whitney p-value	0.664	No significant family gap
Behavior	Joint pass rate (none / A / B / A+B)	0.000 / 0.043 / 0.043 / 0.043	Summed steering does not beat best single
Behavior	Composition gain over best single	0.000	No additive behavioral benefit
Transfer	TruthfulQA local additivity cosine	0.868	High local additivity
Transfer	HellaSwag local additivity cosine	0.919	High local additivity

Table 1: Main quantitative results. We report representation-space, behavior-space, and transfer/locality metrics. Best value for the primary representation metric is highlighted in **bold**.

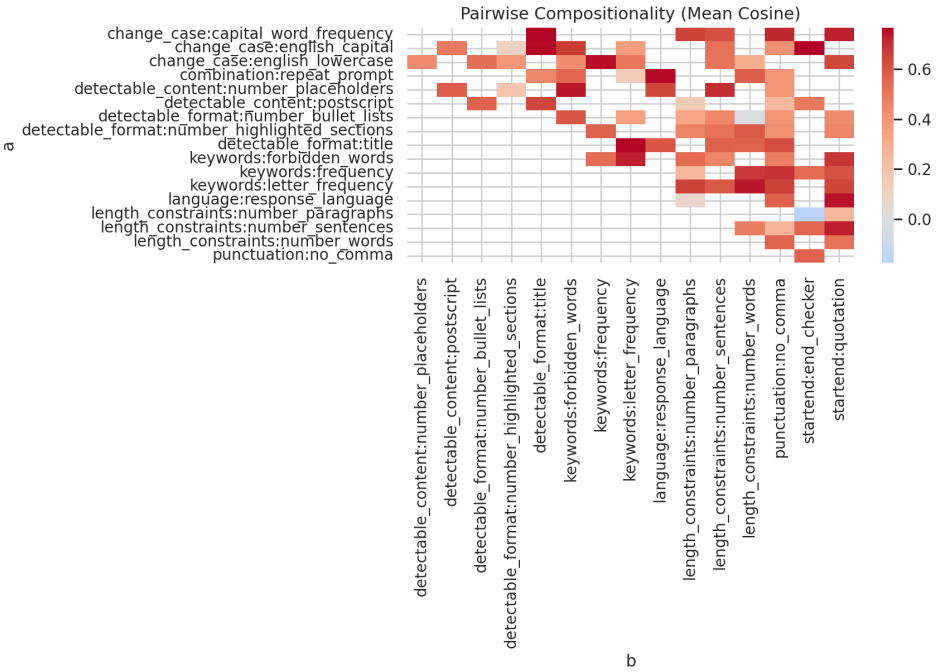

Figure 1: Pairwise compositionality heatmap over instruction-direction pairs. Bright regions indicate high mean cosine alignment; dark regions indicate weak or negative composition. The map shows substantial heterogeneity rather than a uniform linear subspace.

the mean R2-like score is negative (-0.495), which indicates poor reconstruction quality despite moderate directional agreement.

4.2 Pairwise Compositionality Map

As shown in figure 1 and figure 2, the strongest observed pairs include `detectable_format:title + keywords:letter_frequency` (0.767) and `change_case:capital_word_frequency + detectable_format:title` (0.766). The weakest include `length_constraints:number_paragraphs + startend:end_checker` (-0.176) and `detectable_format:number_bullet_lists + length_constraints:number_words` (-0.025). This spread includes negative values, confirming strong pair-specific interference.

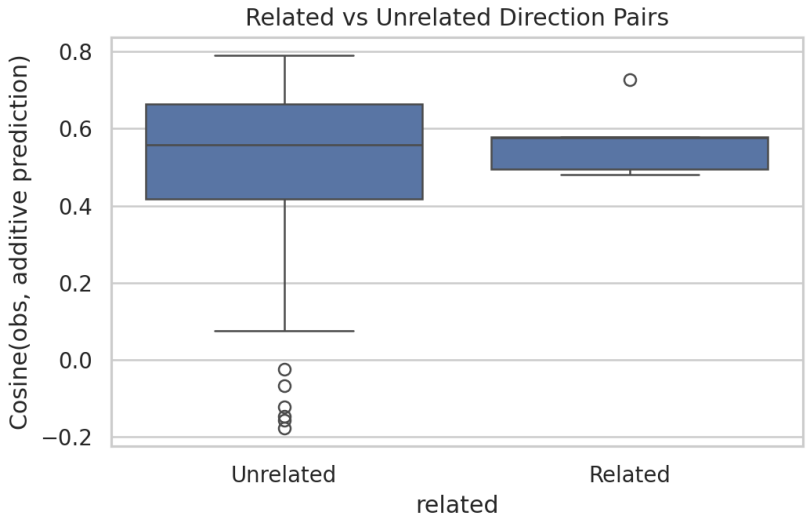

Figure 2: Related vs. unrelated direction-pair cosine alignment. Related pairs have slightly higher mean (0.571 vs. 0.509), but the difference is not significant (Mann-Whitney $p = 0.664$).

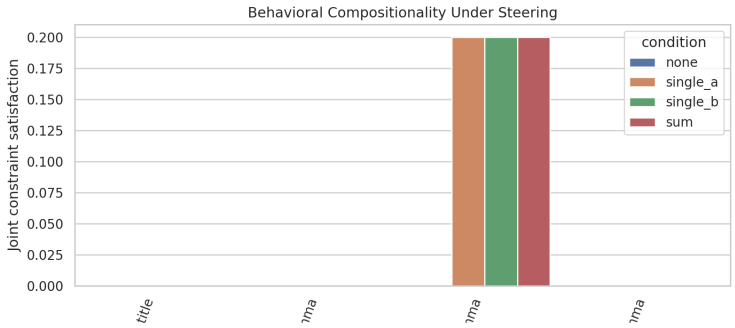

Figure 3: Joint two-constraint satisfaction rates under no steering, single-direction steering, and summed steering. Additive steering (A+B) matches but does not exceed the best single condition.

4.3 Behavioral Compositionality

Behavioral outcomes mirror representation-space fragility. Joint satisfaction is 0.000 with no steering and 0.043 for each of single-A, single-B, and A+B. The composition gain over best single is therefore 0.000, indicating no practical improvement from naive vector summation in this subset.

4.4 Transfer and Locality

Transfer tests show high additivity for local perturbations (TruthfulQA 0.868; HellaSwag 0.919). Combined with weaker full-bundle results on IFEval, this pattern supports a locality interpretation: linear composition appears stronger for small contextual perturbations than for full instruction bundles.

4.5 Reproducibility and Sensitivity

A complete rerun with the same seed reproduces key summary metrics exactly (`summary_run1.json` vs. `summary_run2.json`). This deterministic match reduces concern that the observed non-compositionality is an artifact of random execution variance.

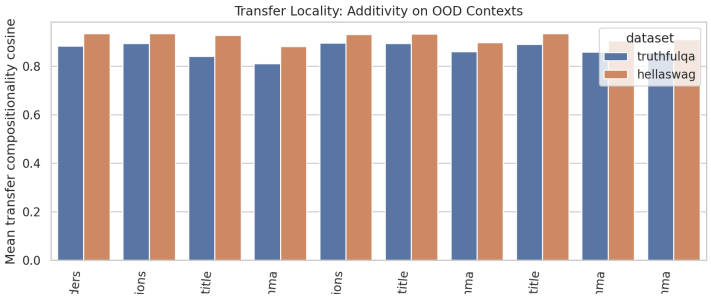

Figure 4: Transfer/locality additivity on external prompts with synthetic instruction appends. Additivity is high (0.868 on TruthfulQA and 0.919 on HellaSwag), suggesting local linear structure persists even when full in-domain composition is uneven.

5 Discussion

Interpretation. Our results support partial compositionality, not a globally coherent linear direction space. Moderate cosine with negative mean R2-like score implies a common failure mode: additive predictions often point in roughly the right direction but miss the correct scale and detailed structure. The behavior-space result is consistent with this mismatch, since summed steering fails to improve over the best single direction.

Why does transfer look better? The transfer/locality experiment applies synthetic instruction appends that act like local perturbations. High transfer cosines can therefore coexist with weak global composition on full constrained-to-base displacements. This aligns with context-dependent steering hypotheses: local tangent-like linearity may hold even when global vector addition fails [Li et al., 2026].

Limitations. Our study uses one model scale (0.5B) and a single intervention layer. Behavioral checkers are heuristic and cover a limited instruction subset. Steering coefficients are fixed rather than pair-specific tuned. Transfer tests use synthetic appends rather than full downstream task execution. Related-vs-unrelated family effects may be underpowered in this sample.

Broader implications. For safety and control applications, our findings argue against assuming universal additivity when composing constraints. Practically, direction composition should be calibrated per pair (and possibly per context), or replaced with adaptive schemes that model local geometry. For interpretability, averaging over direction pairs can hide severe failure pockets that matter operationally.

6 Conclusion

We presented a direction-level study of compositionality in residual-stream steering vectors. Across 20 instruction directions and 85 pairs, we found a mixed regime: representation-space alignment is moderate on average, reconstruction quality is often poor, and behavior-level additive gains are negligible in our evaluated subset.

The key takeaway is simple: residual directions are partially compositional but not uniformly compositional. Assuming a single global linear subspace is too strong for reliable multi-constraint steering.

Future work should (i) tune coefficients and layers per pair, (ii) repeat analyses on larger models, (iii) compare static vectors with context-dependent vector fields, and (iv) evaluate with stricter task-level scorers for long-form instruction following.

References

Jiaqian Li, Yanshu Li, and Kuan-Hao Huang. Steering vector fields for context-aware inference-time control in llms. *arXiv preprint*, 2026. Preprint.

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023. URL https://arxiv.org/abs/2301.05217.

Nina Panickssery, Esin Durmus, Alexander Matt Turner, James Zou, Stefano Ermon, et al. Steering llama 2 via contrastive activation addition. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://arxiv.org/abs/2312.06681.

Daniel Scalena, Gabriele Sarti, and Malvina Nissim. Dynamic activation composition for multi-property language model steering. *arXiv preprint*, 2024. Preprint.

Adam Shai and collaborators. Transformers learn factored representations. *arXiv preprint*, 2026. Preprint.

Alessandro Stolfo, Nora Belrose, Yilun Du, et al. Improving instruction-following through activation steering. In *International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2410.12895.

Jiuding Sun and collaborators. Hypersteer: Hypernetwork-based activation steering for language models. *arXiv preprint*, 2025. Preprint.

Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. Function vectors in large language models. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=AwyxtyMwaG.

Alexander Matt Turner, Ludwig Thiergart, Gavin Leech, David Udell, Juan Vazquez, Ryan Greenblatt, Monte MacDiarmid, et al. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL https://arxiv.org/abs/2308.10248.